

CPE 381 CLASSWORK 09: TIME-INVARIANCE, PERIOD OF A SIGNAL

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 23, 2026 (In-Class)

1. Is the system described by input-output relationship $y(t) = x(t) \cdot \cos(2\pi t)$ a time-invariant system? Show your work. **(10 Points)**

Answer**Time-Variant.**

Check: For input $x_d(t) = x(t - t_0)$, the output is $T[x_d(t)] = x_d(t) \cdot \cos(2\pi t) = x(t - t_0) \cdot \cos(2\pi t)$.

The shifted output is $y(t - t_0) = x(t - t_0) \cdot \cos(2\pi(t - t_0))$.

Since $\cos(2\pi t) \neq \cos(2\pi(t - t_0))$ for all t_0 not an integer, the system is time-varying.

2. Find the fundamental period of $x(t) = \cos(2\pi t) + \sin(4\pi t) + \cos(6\pi t)$. **(10 Points)**

Answer

- $T_1 = 1$ s (for $\cos(2\pi t)$)
- $T_2 = 0.5$ s (for $\sin(4\pi t)$)
- $T_3 = 1/3$ s (for $\cos(6\pi t)$)
- All ratios are rational, hence, commensurable.
- LCM of T_1, T_2, T_3 : since $T_1 = 1$, T must be integer. Check $T = 1$ s: multiple of T_2 and T_3
- **Thus, fundamental period: 1 second**